Rocket Propulsion – Homework Project 2025

Psyche Mission: Trajectory and Propulsion Analysis

Nathan Fagnou

December 9, 2025

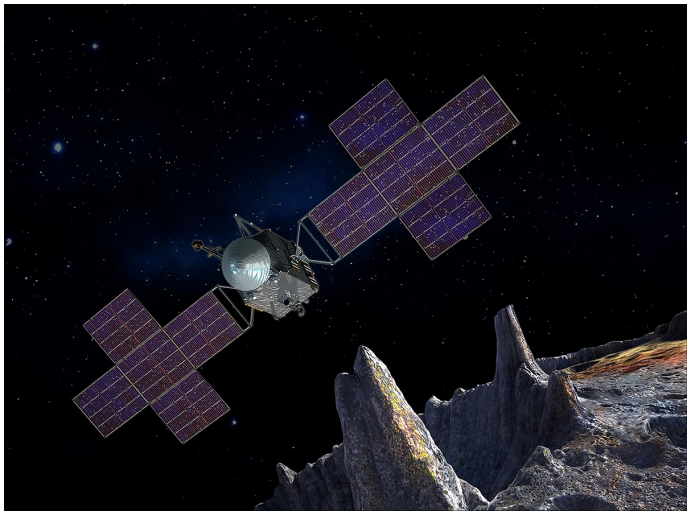


Figure 1: NASA Psyche spacecraft [1]

Abstract

Asteroid (16) Psyche is one of the largest asteroids in the main asteroid belt and is a type M asteroid (for Metal). It was discovered in 1852 by Annibale De Gasparis. It orbits the Sun at roughly 3 AU and has a diameter of about 250 km, making it one of the ten most massive asteroids. What makes Psyche exceptional is its unusual composition: spectral measurements indicate that it is rich in metals, particularly iron and nickel, unlike most asteroids, which are rocky or icy. Because of this, Psyche is often described as the possible exposed core of an early protoplanet whose outer layers were stripped away by violent collisions during the early Solar System.

If this interpretation is correct, 16 Psyche would offer the first direct access to planetary core material, something we cannot observe even inside Earth.

Contents

1 Psyche and Falcon Heavy datasheet **3**
1.1 Psyche spacecraft datasheet [1] 3
1.2 Falcon Heavy datasheet [2] 4

2 Psyche mission and launch profile, ΔV and propellant mass used **5**
2.1 Psyche mission and launch profile 5
2.2 Calculation of needed Δv and propellant mass 5

3 Needed Δv and propellant mass for a direct flight to Psyche asteroid from any Earth's parking orbit, without any trajectory gravity assist (as will be done in real life). **7**
3.1 Direct trajectory 7
3.2 Entering an orbit around the Sun intersecting Psyche, no intermediate slingshots around other bodies. 8

4 Orbital slingshots (gravity assists) **10**

5 The Oberth effect **11**

6 Orbit insertion around (16) Psyche **12**

7 Hall-effect thrusters versus classic thermal rocket engines **14**

8 acknowledgements **15**

1 Psyche and Falcon Heavy datasheet



Figure 2: Falcon Heavy first flight

1.1 Psyche spacecraft datasheet [1]

Table 1: Main characteristics of the Psyche spacecraft

Parameter	Value
Mission target	Asteroid 16 Psyche
Launch date	13 October 2023
Launch vehicle	Falcon Heavy
Launch mass (wet)	2747 kg
Propellant (xenon)	1085 kg
Solar array power (at 1 AU)	4.5 kW
Number of main thrusters	4
Thruster type	SPT-140 Hall-effect thruster
Thruster nominal thrust	0.28 N per thruster
Specific impulse	1800 s
Total impulse capability	8.2×10^6 N s (about 8.2 MN.s)

1.2 Falcon Heavy datasheet [2]

Table 2: Main characteristics of Falcon Heavy

Parameter	Value
Height	70 m
Core / booster diameter	3.7 m
Overall width	12.2 m
Maximal mass	1 420 788 kg
Dry mass	72 300 kg
Payload to Low Earth Orbit (LEO)	63 800 kg
Payload to Geostationary Transfer Orbit (GTO)	26 700 kg
Payload to Mars	16 800 kg
Liftoff mass	1.42×10^6 kg
Propellant	LOX / RP-1
Stages	2.5 (two boosters + center core + upper stage)
Total thrust at liftoff (SL)	15 200 kN
Total thrust in vacuum	16 400 kN
Specific impulse (SL)	282 s (2.77 km/s)
Specific impulse (vacuum)	311 s (3.05 km/s)

Table 3: Falcon Heavy first-stage and booster data

Parameter	Value
Number of boosters	2 side boosters + 1 center core
Engines per core	9 × Merlin 1D (sea-level)
Propellant	LOX / RP-1
Thrust per core (SL)	7600 kN
Thrust per core (vac)	8200 kN
Total thrust (3 cores, SL)	15 200 kN
Total thrust (3 cores, vac)	16 400 kN
Specific impulse (SL)	282 s (2.77 km/s)
Specific impulse (vac)	311 s (3.05 km/s)
Burn time (boosters / center core)	$\sim$ 180 s

Table 4: Falcon Heavy second-stage data

Parameter	Value
Stage type	Second stage
Engine	1 × Merlin 1D Vacuum
Propellant	LOX / RP-1
Thrust (vacuum)	934 kN
Specific impulse (vacuum)	348 s (3.41 km/s)
Burn time	397 s

2 Psyche mission and launch profile, ΔV and propellant mass used

2.1 Psyche mission and launch profile

Nasa approved a mission for Psyche on January 4, 2017 and attributed to SpaceX for a contract of 117 million dollars. The mission profile is the following ;
- Launch on the 13 October 2023 using a falcon heavy rocket from Cape Canaveral's base.
- Verification of the probe's systems during the first 100 days after launch
- Mars gravity assist (May 2026)
- Tests of the experimental DSOC laser communication system (pink points)
- Spacecraft trajectory
- Orbital insertion around Psyche (August 2029)

The spacecraft will only use one of its four thrusters to maintain a certain temperature interval during the mission. The Psyche mission is also the first mission to use **Hall-effect thrusters**. The main goal of the mission is to learn about the process of planet's formation. To make this mission successful, Psyche is equipped with a magnetometer, spectrometers and two cameras to measure magnetic field, presence of chemical elements, etc.

2.2 Calculation of needed Δv and propellant mass

We will use the following hypothesis to determine the Δv :
- The launch is vertical
- The launch is drag free

To escape Earth gravity from the ground an object must have the following escape velocity ;

$$\text{es} = \sqrt{\frac{2GM}{R}}$$

Numerical application :

$$\boxed{v_{\text{es}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.378 \times 10^{6}}} = 11.2\, km.s^{-1}}$$

Therefore, the needed ΔV must be 11.2 $km.s^{-1}$.

From Falcon Heavy's datasheet ke know that the maximal wet mass is 1.420.788 kg (thus including payload for LEO), the dry mass is 72.300 kg and payload for LEO is 63.800 kg. To find m_0 we have to add the mass of Psyche, thus ;

$$m_0 = m_{max} - m_{payload,LEO} - m_{dry\,mass} + m_{psyche}$$

where :

$$m_{psyche} = m_{launch\,mass} + m_{propellant}$$

Numerical application :

$$m_0 = 1.420.788 - 63.800 - 72.300 + 2747 + 1085 = 1.288.520\, kg$$

Now for the propellant mass, we will use the Tsiolkovski's equation ;

$$\Delta v = c\, \ln\left(\frac{m_0}{m_{\text{f}}}\right)$$

We can wewrite it using the Specific Impulse and g ;

$$\Delta v = I_s \, g \ln\left(\frac{m_0}{m_\mathrm{f}}\right)$$

We then isolate m_0:

$$m_0 = m_f \, \exp\left(\frac{\Delta v}{I_{sp} \, g}\right)$$

Knowing $m_p = m_0 - m_f$;

$$m_p = m_0(1 - \, \exp\left(\frac{\Delta v}{I_{sp} \, g}\right)$$

We know want to find $\dot{m}_{tot}$ knowing :

$$\dot{m}_\mathrm{tot} = \sum_i \frac{F_i}{I_{sp,i} \, g}$$

We can then replace $\dot{m}_{tot}$:

$$I_{sp,\mathrm{tot}} = \frac{\sum_i F_i}{\sum_i \frac{F_i}{I_{sp,i}}}$$

Numerical application for $I_{sp,\mathrm{tot}}$:

$$I_{sp,\mathrm{tot}} = \frac{7600.10^3 + 7600.10^3 + 934.10^3}{\frac{7600.10^3}{282} + \frac{7600.10^3}{282} + \frac{934.10^3}{348}} = 285s$$

We can know calculate m_p :

$$\boxed{m_p = 1288520(1 - \, \exp\left(\frac{-11,2.10^3}{285 * 9,81}\right) \approx 1265 \, \mathrm{Tons}}$$

The result we find is inferior to the maximum ergols mass cited in Destination-Orbite's article about Falcon Heavy which seems coherent [3].

3 Needed Δv and propellant mass for a direct flight to Psyche asteroid from any Earth's parking orbit, without any trajectory gravity assist (as will be done in real life).

3.1 Direct trajectory

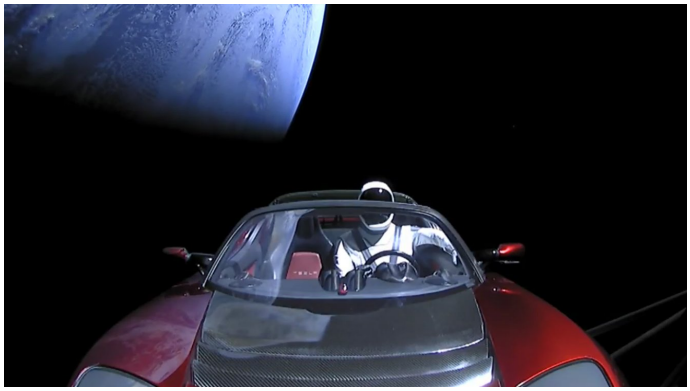


Figure 3: Tesla's roadster direct trajectory to (16) Psyche

We know consider a direct trajectory to (16) Psyche, we found at the previous question that the need velocity to escape earth attraction was :

$$v_{es,earth} = 11,2\,km.s^{-1}$$

But now we also have to consider the sun's attraction. Knowing that the distance from Earth to the Sun is equal to one Astronomical Unit :

$$1\,AU \approx 1,496.10^{11}\,m$$

To escape sun's attraction, we must have the following escape velocity :

$$v_{es,sun} = \sqrt{\frac{2 * G * M_{sun}}{AU + h}}\, where\, h = R_{earth}$$

where h = R_{earth}, but $R_{earth} << 1AU$, thus We obtain the following expression :

$$v_{es,sun} = \sqrt{\frac{2 * G * M_{sun}}{AU}}$$

Numerical application :

$$v_{es,sun} \approx \sqrt{\frac{2 * 6,67.10^{-11} * 1,989.10^{30}}{1,496.10^{11}}} \approx 4,21.10^{5}\,m.s^{-1} \approx 42,1\,km.s^{-1}$$

Since Earth orbits the Sun at an orbital speed of about $29.8km.s^{-1}$, or true escape velocity is the following :

$$v_{es,sun,true} = v_{es,sun} - v_{orb,earth}$$

Numerical application :

$$\boxed{v_{es,sun,true} = 42.1 - 29.8 = 12,3km.s^{-1}}$$

Using the expression of m_p from question one, we have the following expression in function of $v_{es,sun}$:

$$m_p = m_0(1 - \exp\left(\frac{v_{es,sun,true}}{I_{sp}\, g}\right)$$

Numerical application :

$$\boxed{m_p = 1288520(1 - \exp\left(\frac{-12,3.10^3}{285 * 9,81}\right) \approx 1273\,\text{Tons}}$$

The distance between the sun and (16) Psyche is between 2.53 et 3.32 AU [1] so the distance between Earth and (16) Psyche is between 300 and 600 million kilometers. So if we consider a constant speed of 42.1 $km.s^{-1}$, we can fin a interval for the needed time to reach (16) Psyche :

$$\frac{300.10^6}{42,1} \leq T_{travel} \leq \frac{600.10^6}{42,1}$$

Numerical application :

$$7,12.10^6 s \leq T_{travel} \leq 14,3.10^5 s$$

So converting in days :

$$82\,days \leq T_{travel} \leq 164\,days$$

3.2 Entering an orbit around the Sun intersecting Psyche, no intermediate slingshots around other bodies.



Figure 4: Toy Story meme

In this case, we will use the **Hohmann transfer orbit** [10] between Earth's orbit and (16) Psyche's orbit because it reduces the amount of propellant needed in our case (2 maneuvers trajectory).

Since Psyche (the probe satellite) can be considered as a small body compared to earth or the Sun, its total energy is the following ;

$$E = \frac{mv^2}{2} - \frac{GMm}{r} = \frac{-GMm}{2a}$$

By solving this equation, we obtained the **vis-viva equation** :

$$v^2 = \mu(\frac{2}{r} - \frac{1}{a})$$

v is the speed of Psyche. $\mu = GM$ is the standard gravitational parameter, M the mass of the sun. r is the distance of Psyche to the sun/(16) Psyche. a is the semi-major axis of Psyche's orbit. The

two Δv needed are the following :

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}(\sqrt{\frac{2r_2}{r_1+r_2}} - 1)$$

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2}}(1 - \sqrt{\frac{2r_1}{r_1+r_2}})$$

where $r_1 = 1\,AU$ and $r_2 \approx 2.92\,AU$ (we will take the mean of between 2.53 et 3.32 AU) Numerical application ;

$$\Delta v_1 = \sqrt{\frac{6,67.10^{-11} * 1,989.10^{30}}{1,496.10^{11}}}(\sqrt{\frac{2 * 2,92 * 1,496.10^{11}}{1,496.10^{11} + 2,92 * 1,496.10^{11}}} - 1) \approx 6.57\,km.s^{-1}$$

$$\Delta v_2 = \sqrt{\frac{6,67.10^{-11} * 1,989.10^{30}}{2,92 * 1,496.10^{11}}}(1 - \sqrt{\frac{2 * 1,496.10^{11}}{1,496.10^{11} + 2,92 * 1,496.10^{11}}}) \approx 4.98\,km.s^{-1}$$

So finally the total Δv is :

$$\Delta v_{total} = \Delta v_1 + \Delta v_2$$

$$\boxed{\Delta v_{total} = 11,55\,km.s^{-1}}$$

Now for the propellant mass needed, we will still use Tsiolkovski's equation :

$$m_p = m_0(1 - \exp\left(\frac{-\Delta v_{total}}{I_{sp}\,g}\right))$$

Numerical application :

$$\boxed{m_p = 1288520(1 - \exp\left(\frac{-11,55.10^3}{285 * 9.81}\right)) \approx 1268\,Tons}$$

Using Kepler's third law, we can find the time taken to transfer between the orbits :

$$t_H = \pi\sqrt{\frac{(r_1+r_2)^3}{8\mu}}$$

$$\boxed{t_H \approx 501\,days}$$

The Gap between the two trajectories it clearly not negigible, but this trajectory is still way more time effective than the actual mission (which is around 6 years).

4 Orbital slingshots (gravity assists)

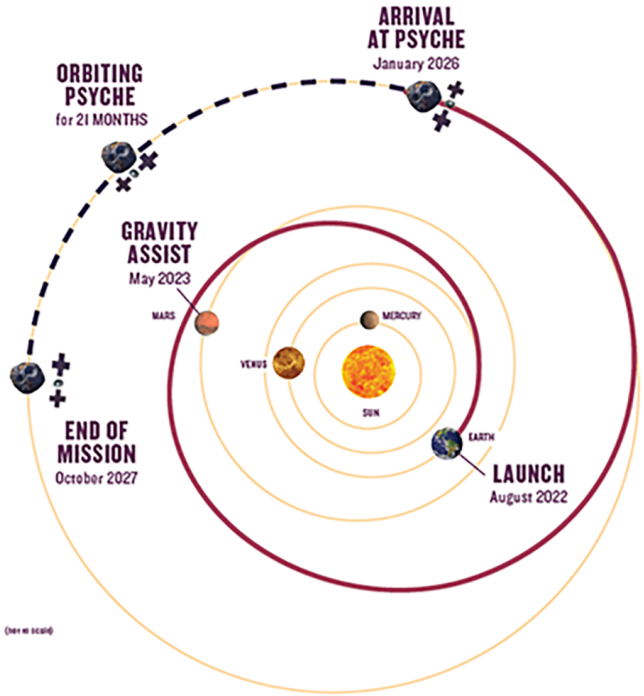


Figure 5: Gravity maneuvers of Psyche [4]

A gravity assist (also called orbital slingshot) is a technique where a spacecraft flies close to a planet and uses its gravity field to change its velocity and trajectory, without propellant use. The idea is that the spacecraft exchanges energy and momentum with the planet.

In the reference frame of the planet, the spacecraft comes in on a hyperbolic trajectory and leaves with almost the same speed, only the direction has changed. But in the heliocentric frame (around the Sun), the planet itself is moving. Because of this, after the flyby the spacecraft can have a higher or a lower heliocentric velocity, depending if it passes "in front of" or "behind" the planet relative to its motion.

In practice, mission designers use gravity assists to:

- increase the heliocentric speed of the spacecraft and reach a more distant orbit (for example outer planets or the asteroid belt);
- or decrease its speed to fall toward the inner Solar System;
- change the inclination or direction of the trajectory without paying the full Δv with rocket engines.

For the Psyche mission, the spacecraft will use a Mars gravity assist in 2026. By flying through the gravity field of Mars and leaving it in the correct direction, Psyche increases its speed and adjusts its orbit so that it can intersect the orbit of asteroid 16 Psyche, while saving a significant amount of chemical or electric propellant compared to a direct impulsive transfer [7].

5 The Oberth effect



Figure 6: Escape of Gargantua, Interstellar

The Oberth effect describes the fact that applying thrust when a spacecraft is deep inside a gravitational well, and therefore moving at high velocity, produces a much larger increase in kinetic energy than the same thrust applied farther away. The difference between a gravity assist and the Oberth effect is that gravity assists do not require any propellant, while the Oberth effect explicitly comes from firing the engine at the right moment.

This means that the spacecraft sweeps equal areas in equal times, and therefore its velocity is highest at periapsis and lowest at apoapsis. Because of this, performing a burn at periapsis maximises the work done by the engine and the resulting gain in orbital energy. The Oberth effect is therefore strongest when the spacecraft is at its closest point to the central body.

This effect is well described in Interstellar when Cooper and his crew use the gravitaionnal well of gargantua to gain velocity and, when reaching maximum velocity, use every thrusters to escape Gargantua's attraction.

6 Orbit insertion around (16) Psyche

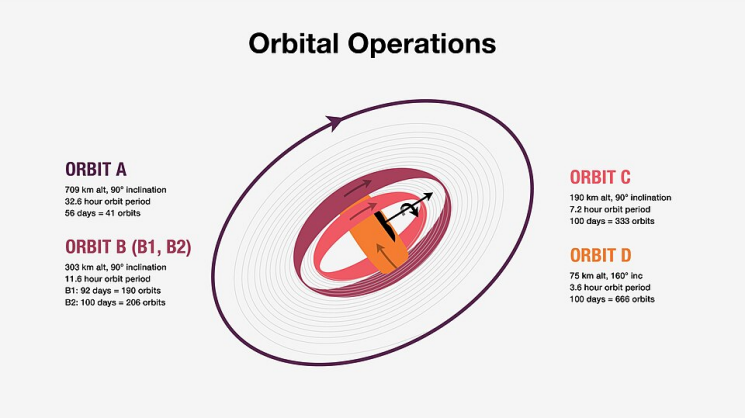


Figure 7: Orbital operations of Psyche [1]

We consider here a circular orbit similar to Orbit A from the real mission profile, at an altitude

$$h \approx 709\,\text{km}$$

above the surface of 16 Psyche.

The effective radius of Psyche is about

$$R \approx 126.5\,\text{km}\,[5],$$

so the orbital radius is

$$r = R + h \approx 126.5 + 709 \approx 836\,\text{km}.$$

Recent analyses give a gravitational parameter for Psyche of

$$\mu = GM = 6.67.10^{-11} * 1.7.10^{19} \approx 1,1.10^{9}\, m^3.s^{-2}.$$

The circular orbital velocity at this altitude is then

$$v_{\text{orb}} = \sqrt{\frac{\mu}{r}} = \sqrt{\frac{1,1.10^9}{836.10^3}} = 36\,\text{m/s}.$$

If we assume that, at arrival, Psyche has already reduced its relative velocity with respect to the asteroid to a very small value (typical for a solar electric propulsion rendezvous), the order of magnitude of the Δv needed to be captured in this circular orbit is

$$\Delta v_{\text{capture}} \approx v_{\text{orb}} \approx 36\,\text{m/s}.$$

Now we can estimate the propellant mass needed on board the Psyche spacecraft to provide this Δv using its Hall-effect thrusters. We approximate:

- initial wet mass of the spacecraft (at launch): $m_0 \approx 2747\,\text{kg}$,
- specific impulse of the SPT-140 thrusters: $I_{sp} \approx 1800\,\text{s}$,
- standard gravity: $g = 9.81\,\text{m/s}^2$.

Using Tsiolkovski's equation in the form

$$\Delta v = I_s\, g \ln\left(\frac{m_0}{m_f}\right),$$

we can isolate the final mass m_f and deduce the propellant mass:

$$m_{\mathrm{p}} = m_0 - m_f = m_0 \left(1 - \exp\left(-\frac{\Delta v_{\mathrm{capture}}}{I_s g}\right)\right).$$

Numerical application:

$$\frac{\Delta v_{\mathrm{capture}}}{I_s g} = \frac{36}{1800 \times 9.81} \approx 2.0 \times 10^{-3},$$

so

$$m_{\mathrm{p}} \approx 2747 \left(1 - e^{-2.0\times10^{-3}}\right) \approx 5.489\,\mathrm{kg}.$$

Therefore, the Δv needed to slow down and enter a low orbit around Psyche is of the order of

$$\boxed{\Delta v_{\mathrm{capture}} \approx 36\,\mathrm{m/s}}$$

and requires only about

$$\boxed{m_{\mathrm{p}} \approx 5.489\,\mathrm{kg}}$$

of xenon if done with the high-I_{sp} Hall-effect thrusters. Compared to the total xenon mass on board (about 915 kg), this is only about 0.6%, so the increase of launch mass due to orbit insertion propellant is very small.

7 Hall-effect thrusters versus classic thermal rocket engines

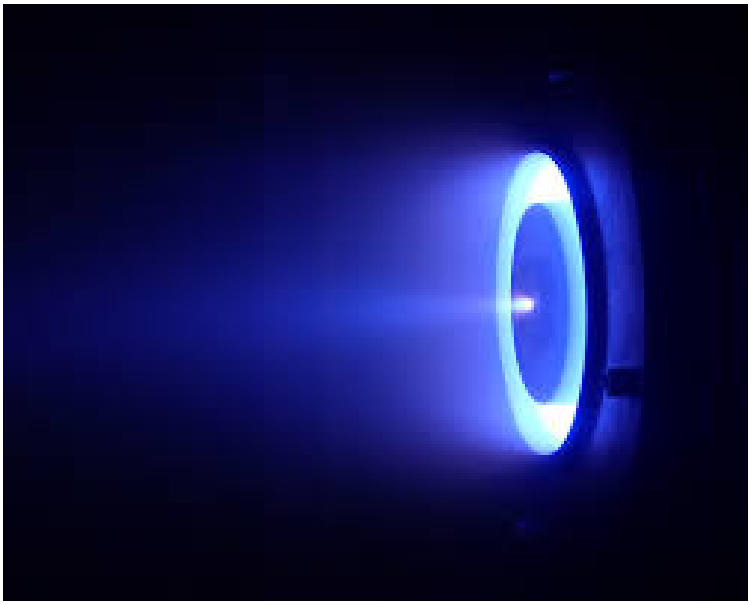


Figure 8: Hall effect thruster operating with Krypton [9]

Hall-effect thrusters, like the SPT-140 used on Psyche, are electric propulsion devices. They use electrical power from the solar arrays to ionise a neutral gas (xenon) and accelerate the ions with an electric field, while a magnetic field confines the electrons [5]. The result is a plasma jet with a very high exhaust velocity, and therefore a very high specific impulse compared to chemical rockets.

For Psyche's SPT-140, the nominal characteristics are:

- thrust $\approx 0.28\,\mathrm{N}$ per thruster,
- specific impulse $I_{sp} \approx 1800\,\mathrm{s}$,
- input power up to about $4.5\,\mathrm{kW}$.

In comparison, classic chemical rocket engines (thermal engines using combustion of LOX/RP-1, LOX/LH_2, etc.) typically have:

- much higher thrust (kN to MN range),
- but a much lower specific impulse, often around $300\,\mathrm{s}$ for kerosene engines and up to $450\,\mathrm{s}$ for the best hydrogen engines.

The main advantages of Hall-effect thrusters in deep space are:

- **Much higher propellant efficiency:** with $I_{sp} \approx 1800$ s, the same total Δv can be obtained with several times less propellant mass than with chemical propulsion. This is very important for interplanetary missions where launch mass is limited.
- **Continuous low thrust:** the thrusters can operate for thousands of hours at low thrust. This allows slow but very efficient changes of orbit (spiral out from Earth orbit, rendezvous with an asteroid, changes of science orbit around the target, etc.).
- **Good use of solar power:** far from Earth, solar panels can still provide electrical power, which is converted into kinetic energy of the ions instead of using chemical energy stored in propellants.

The disadvantages are mainly:

- the thrust is very low, so they cannot be used for launch from the ground or for fast impulsive maneuvers;
- the mission design must accept long thrust durations (months or years) to build up the required total Δv.

For Psyche, Hall-effect thrusters are an excellent solution: they allow the spacecraft to accumulate a large total Δv during the cruise and to perform all the orbital changes around the asteroid, while keeping the xenon mass (and so the launch mass) at a reasonable level.

8 acknowledgements

I would like to thank Mister GARCIA-MARTIN Pablo for this homework assignment. Honestly, it was quite challenging and more interesting than conventional exams or traditional homework assignments.I hope you enjoyed reading mine and hopefully the memes did not bother you !

Sources

1 Psyche Spacecraft (Wikipedia) — https://en.wikipedia.org/wiki/Psyche_(spacecraft)

2 SpaceX Falcon Heavy — https://www.spacex.com/vehicles/falcon-heavy/

3 La fusée américaine Falcon Heavy — https://destination-orbite.net/astronautique/compendium/falcon-heavy

4 article photo - https://medium.com/the-nasa-psyche-mission-journey-to-a-metal-world/psyche-is-an-sep-mission-whats-that-mean-f56f7f0c8a89

5 (16) Psyche - https://fr.wikipedia.org/wiki/(16)_Psych%C3%A9

6 Psyche Fact Sheet, NASA/JPL — https://d2pn8kiwq2w21t.cloudfront.net/documents/Psyche_Fact_Sheet_9.4.23.pdf

7 Psyche Mission Timeline (ASU/NASA) — https://psyche.ssl.berkeley.edu/timeline/

8 Mass, Density, and Radius of Asteroid (16) Psyche from High-precision Astrometry, AJ 168, 21 (2024).

9 Illustration of a Hall effect Thruster - https://projects-web.engr.colostate.edu/ionstand/research/research.php

10 Hohmann transfer orbit (Wikipedia) - https://en.wikipedia.org/wiki/Hohmann_transfer_orbit